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Pages: 4

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 scheme)

Course Code: EET201

Course Name: CIRCUITS AND NETWORKS

Max. Marks: 100

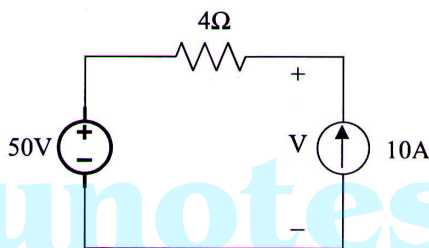
Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

- 1 State and explain Reciprocity theorem using an example. (3)
- 2 For the circuit given below, find the voltage 'V' across the 10A source, using superposition principle. (3)



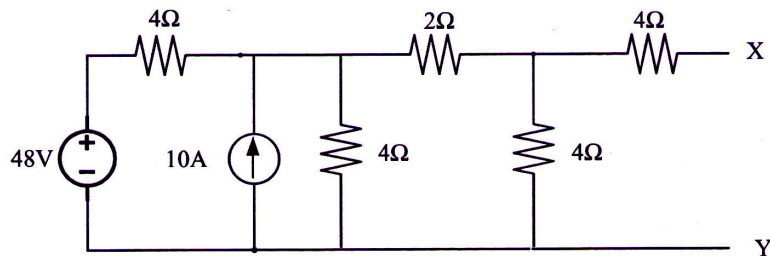
- 3 Derive an expression for the current in a series RL circuit connected to a DC source. (3)
- 4 Define time constant of a circuit. A series RC circuit is connected to a DC source of 10V at time $t = 0$. Find the voltage across the capacitor at $t = 1s$, if the time constant of the circuit is 2 seconds. (3)
- 5 Define transfer function of a network. Derive the transfer function of a series RLC circuit by taking the current through the resistor as the output. (3)
- 6 Explain the use of dot convention in the analysis of coupled circuits. (3)
- 7 Determine the current through the neutral wire in a three phase 4 wire system, if the phase currents of the load are $I_R = 10\angle -30^\circ A$, $I_Y = 5\angle -50^\circ A$ and $I_B = 5\angle 50^\circ A$. (3)
- 8 A series RLC circuit with $R = 10\Omega$, $L = 2H$ and $C = 40\mu F$ is connected to a variable frequency AC supply. Determine the frequency of the supply, for which the phase angle between the circuit current and supply voltage is zero. (3)
- 9 Define h parameters of a two-port network. Why are they called hybrid parameters? (3)
- 10 Obtain Y parameters of a two-port network whose Z parameters are given by $z_{11} = 4\Omega$, $z_{12} = 2\Omega$, $z_{21} = 3\Omega$ and $z_{22} = 4\Omega$. (3)

PART B

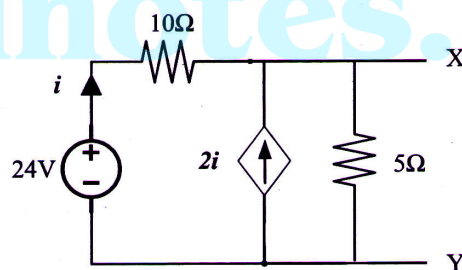
Answer any one full question from each module. Each question carries 14 marks

Module 1

- 11 For the circuit given below,
- Determine the Thevenin's equivalent circuit across the terminals X and Y. (10)
 - Determine the value of resistance to be connected across X and Y so that maximum power is transferred to it. Also, calculate the maximum power transferred. (4)

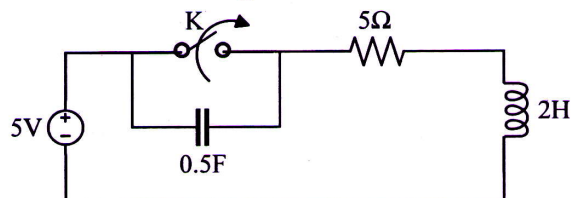


- 12 For the circuit given below,
- Find the Norton's equivalent circuit across the terminals X and Y. (10)
 - If a 10Ω resistor is connected across the terminals X and Y, find the power dissipated in it. (4)



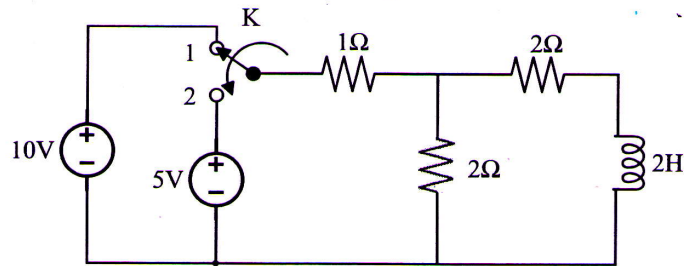
Module 2

- 13 The switch K in the circuit given below has been in the closed position for a long time and steady state condition is reached. At $t = 0$, the switch is opened. Find the expression for the current through the inductor for $t > 0$. (14)



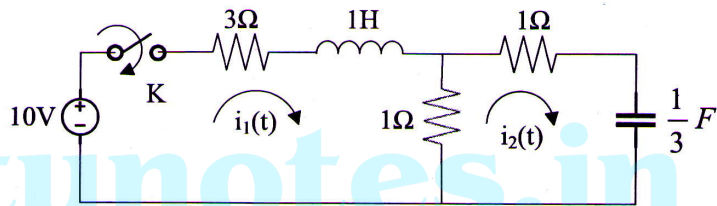
0800EET201122101

- 14 The switch K in the circuit given below was initially at position 1 and the circuit has been at steady state condition. At time $t = 0$, the switch is moved to position 2. Find the expression for the current through the inductor for $t > 0$. (14)

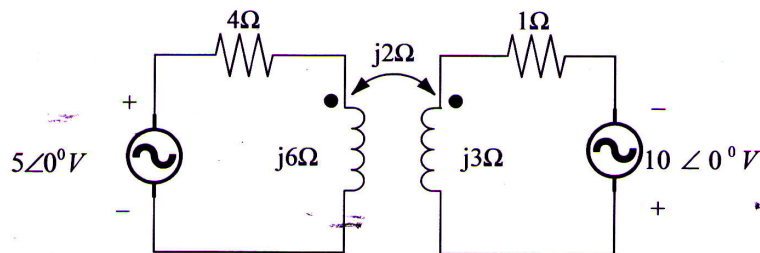


Module 3

- 15 In the network given below, the inductor and capacitor are initially relaxed. The switch K is closed at time $t = 0$.
 a) Model the circuit in s-domain for $t > 0$ (4)
 b) Using mesh analysis, determine the expression for the current through the inductor for $t > 0$. (10)



- 16 For the circuit given below,
 a) Find the steady state current through the 1Ω and 4Ω resistors. (8)
 b) Obtain the conductively coupled equivalent circuit. (6)



Module 4

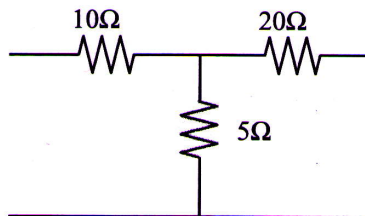
- 17 A 400V, three-phase balanced supply feeds a delta-connected load having phase impedances $Z_{RY} = 40\angle 30^\circ \Omega$, $Z_{YB} = 50\angle 0^\circ \Omega$ and $Z_{BR} = 40\angle -30^\circ \Omega$. Determine the following;
 (i) Phase currents (4)
 (ii) Line currents (6)
 (iii) Active and reactive power delivered to the load (4)

0800EET201122101

- 18 A resistor, a capacitor and an inductor are connected in series with a 230 V, variable frequency AC source. When the supply frequency is varied to 25Hz, the circuit offered a minimum impedance of 50Ω . If the inductance of the circuit is $1H$, determine the following;
- (i) Resistance and capacitance of the circuit. (4)
 - (ii) The voltage across the capacitor. (2)
 - (iii) The supply frequencies at which the power dissipated in the resistor is half that of at 25Hz. (4)
 - (iv) Q factor at resonance and bandwidth of the circuit. (4)

Module 5

- 19 a) Determine the h parameters of the following network. (8)



- b) Derive the condition for symmetry and reciprocity of a two-port network in terms of transmission parameters. (6)
- 20 a) Show that Y parameters of two parallel connected two port networks is equal to the sum of their individual Y parameters. (6)
- b) A two-port network is represented by the following network equations. (8)

$$V_1 = 4I_1 + 2I_2$$

$$V_2 = 2I_1 + 6I_2$$

Determine the equivalent π network.
